Quantum Field Theory

First Exercise Series

Four-Dimensional Lattice Path Propagator

Complete Derivation and Continuum Limit

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Problem

Consider a four-dimensional hypercubic Euclidean lattice with lattice spacing Δ. A path moves only between nearest-neighbour lattice sites. For a path containing L links, assign the weight

$$\mathrm{e}^{-m_0 L}.$$

The lattice propagator is

$$G_\Delta(x) = \sum_{\text{paths } 0\to x} \mathrm{e}^{-m_0 L(\text{path})}. \tag{1}$$

The goal is to compute this path sum and determine how m_0 must depend on Δ in order to obtain a nontrivial continuum limit.

1 Lattice Sites and Allowed Steps

A lattice site is labelled by an integer four-vector

$$n = (n_1, n_2, n_3, n_4), \qquad n_\mu \in \mathbb{Z}. \tag{2}$$

Its physical coordinate is

$$x_\mu = \Delta n_\mu, \qquad x = \Delta n. \tag{3}$$

The vector n is dimensionless, while x has dimensions of length.

At every site there are two directions along each of the four axes,

$$+\hat{e}_\mu, \qquad -\hat{e}_\mu, \qquad \mu = 1, 2, 3, 4.$$

Therefore there are

$$2 \times 4 = 8$$

allowed nearest-neighbour steps at every site.

2 Grouping Paths by Their Number of Links

Let $C_L(n)$ be the number of paths with exactly L steps from 0 to n.

All such paths have the same weight $\mathrm{e}^{-m_0 L}$, hence

$$\boxed{G_\Delta(n) = \sum_{L=0}^{\infty} \mathrm{e}^{-m_0 L} C_L(n).} \tag{4}$$

The initial condition is

$$\boxed{C_0(n) = \delta_{n,0}.} \tag{5}$$

With zero steps, the walker is at the origin only.

3 Long Paths and the Critical Value

An individual path of length L has weight $\mathrm{e}^{-m_0 L}$.

However, if the endpoint is unrestricted, there are 8^L possible step sequences. This does not mean that 8^L paths arrive at a fixed endpoint; for fixed n,

$$C_L(n) \leq 8^L.$$

Summing over all endpoints gives

$$\sum_n C_L(n) = 8^L. \tag{6}$$

Thus the total weight of all unrestricted paths of length L scales as

$$W_L \sim 8^L \mathrm{e}^{-m_0 L} = \left(8\mathrm{e}^{-m_0}\right)^L = \mathrm{e}^{-(m_0 - \ln 8)L}. \tag{7}$$

Therefore,

$$m_0 > \ln 8 \quad \Longrightarrow \quad \text{long paths are exponentially suppressed,} \tag{8}$$

$$m_0 = \ln 8 \quad \Longrightarrow \quad \text{critical boundary of convergence,} \tag{9}$$

$$m_0 < \ln 8 \quad \Longrightarrow \quad \text{the path sum diverges.} \tag{10}$$

For the continuum limit, the critical value is approached from above:

$$m_0(\Delta) \to \ln 8^+. \tag{11}$$

4 Recursion Relation for Path Counting

To arrive at n after $L+1$ steps, the final step must come from one of the eight nearest neighbours of n.

Hence

$$\boxed{C_{L+1}(n) = \sum_{\mu=1}^{4} \left[C_L(n - \hat{e}_\mu) + C_L(n + \hat{e}_\mu)\right].} \tag{12}$$

5 Fourier Transform on the Lattice

We Fourier transform with respect to the lattice coordinate n, rather than the path length L, because a lattice shift becomes multiplication by a phase in Fourier space.

Define

$$\boxed{\widetilde{C}_L(k) = \sum_{n \in \mathbb{Z}^4} \mathrm{e}^{-ik\cdot n} C_L(n).} \tag{13}$$

The inverse transform is

$$\boxed{C_L(n) = \int_{[-\pi,\pi]^4} \frac{\mathrm{d}^4 k}{(2\pi)^4}\, \mathrm{e}^{ik\cdot n} \widetilde{C}_L(k).} \tag{14}$$

Here

$$k \cdot n = \sum_{\mu=1}^{4} k_\mu n_\mu.$$

Since n_μ is an integer,

$$\mathrm{e}^{i(k_\mu + 2\pi)n_\mu} = \mathrm{e}^{ik_\mu n_\mu},$$

so it is sufficient to take

$$-\pi \leq k_\mu \leq \pi.$$

6 Initial Condition in Fourier Space

At $L = 0$, insert $C_0(n) = \delta_{n,0}$ into the Fourier-transform definition:

$$\widetilde{C}_0(k) = \sum_n \mathrm{e}^{-ik\cdot n} C_0(n) \tag{15}$$

$$= \sum_n \mathrm{e}^{-ik\cdot n} \delta_{n,0} \tag{16}$$

$$= \mathrm{e}^{-ik\cdot 0} = 1. \tag{17}$$

Thus

$$\boxed{\widetilde{C}_0(k) = 1.} \tag{18}$$

The Kronecker delta selects only the origin term.

7 Solving the Recursion Relation in Fourier Space

Fourier transforming Eq. (12), use the changes of variables $r = n - \hat{e}_\mu$ and $r = n + \hat{e}_\mu$, respectively:

$$\sum_n \mathrm{e}^{-ik\cdot n} C_L(n - \hat{e}_\mu) = \mathrm{e}^{-ik_\mu} \widetilde{C}_L(k), \tag{19}$$

$$\sum_n \mathrm{e}^{-ik\cdot n} C_L(n + \hat{e}_\mu) = \mathrm{e}^{ik_\mu} \widetilde{C}_L(k). \tag{20}$$

Therefore,

$$\widetilde{C}_{L+1}(k) = \sum_{\mu=1}^{4} \left(\mathrm{e}^{-ik_\mu} + \mathrm{e}^{ik_\mu} \right) \widetilde{C}_L(k) \tag{21}$$

$$= \left[2 \sum_{\mu=1}^{4} \cos k_\mu \right] \widetilde{C}_L(k). \tag{22}$$

We used

$$\mathrm{e}^{i\theta} + \mathrm{e}^{-i\theta} = 2\cos\theta.$$

The factor

$$2 \sum_\mu \cos k_\mu$$

is the one-step hopping factor in Fourier space.

8 Solution for $C_L(n)$

Define

$$F(k) = 2 \sum_{\mu=1}^{4} \cos k_\mu. \tag{23}$$

Since $\widetilde{C}_0(k) = 1$, Eq. (22) gives

$$\boxed{\widetilde{C}_L(k) = F(k)^L = \left[2 \sum_{\mu=1}^{4} \cos k_\mu \right]^L.} \tag{24}$$

Applying the inverse transform,

$$\boxed{C_L(n) = \int_{[-\pi,\pi]^4} \frac{\mathrm{d}^4k}{(2\pi)^4}\, \mathrm{e}^{ik\cdot n} \left[2\sum_{\mu=1}^{4} \cos k_\mu\right]^L.} \tag{25}$$

9 Exact Lattice Propagator

Insert Eq. (25) into Eq. (4):

$$G_\Delta(n) = \int_{[-\pi,\pi]^4} \frac{\mathrm{d}^4k}{(2\pi)^4}\, \mathrm{e}^{ik\cdot n} \sum_{L=0}^{\infty} \left[2\mathrm{e}^{-m_0} \sum_{\mu=1}^{4} \cos k_\mu\right]^L. \tag{26}$$

The sum over L is geometric:

$$\sum_{L=0}^{\infty} r^L = \frac{1}{1-r}, \qquad |r| < 1. \tag{27}$$

Hence

$$\boxed{G_\Delta(n) = \int_{[-\pi,\pi]^4} \frac{\mathrm{d}^4k}{(2\pi)^4}\, \frac{\mathrm{e}^{ik\cdot n}}{1 - 2\mathrm{e}^{-m_0} \sum_{\mu=1}^{4} \cos k_\mu}.} \tag{28}$$

10 Convergence Condition

The largest value of

$$2\sum_{\mu} \cos k_\mu$$

is at $k = 0$:

$$2\sum_{\mu=1}^{4} \cos 0 = 8. \tag{29}$$

Thus convergence requires

$$8\mathrm{e}^{-m_0} < 1, \tag{30}$$

or

$$\boxed{m_0 > \ln 8.} \tag{31}$$

11 Continuum Variables

Take $\Delta \to 0$ while keeping $x = \Delta n$ fixed.
The lattice plane wave is $\mathrm{e}^{ik\cdot n}$, while the continuum plane wave is $\mathrm{e}^{ip\cdot x}$.
Matching their phases gives

$$k \cdot n = p \cdot x = p \cdot (\Delta n), \tag{32}$$

and hence

$$\boxed{k = \Delta p.} \tag{33}$$

Therefore

$$\mathrm{e}^{ik\cdot n} = \mathrm{e}^{ip\cdot x}, \qquad \mathrm{d}^4k = \Delta^4 \mathrm{d}^4p. \tag{34}$$

Equation (28) becomes

$$G_\Delta(x) = \int_{[-\pi/\Delta,\pi/\Delta]^4} \frac{\Delta^4 \mathrm{d}^4 p}{(2\pi)^4} \frac{\mathrm{e}^{ip\cdot x}}{1 - 2\mathrm{e}^{-m_0} \sum_{\mu=1}^{4} \cos(\Delta p_\mu)}. \tag{35}$$

12 Small-Δ Expansion

For fixed physical momentum p,

$$\cos(\Delta p_\mu) = 1 - \frac{\Delta^2 p_\mu^2}{2} + O(\Delta^4 p_\mu^4). \tag{36}$$

With

$$p^2 = \sum_{\mu=1}^{4} p_\mu^2,$$

this gives

$$2\sum_{\mu=1}^{4} \cos(\Delta p_\mu) = 8 - \Delta^2 p^2 + O(\Delta^4 p^4). \tag{37}$$

Hence the denominator is

$$D_\Delta(p) = 1 - 2\mathrm{e}^{-m_0} \sum_{\mu=1}^{4} \cos(\Delta p_\mu) \tag{38}$$

$$= 1 - 8\mathrm{e}^{-m_0} + \mathrm{e}^{-m_0}\Delta^2 p^2 + O(\Delta^4 p^4). \tag{39}$$

13 Why m_0 Must Be Tuned

If $m_0 > \ln 8$ is held fixed, then

$$1 - 8\mathrm{e}^{-m_0}$$

is a nonzero constant, while the p^2 term vanishes as Δ^2.

The continuum limit would then lose the momentum dependence needed for a propagator of the form

$$\frac{1}{p^2 + m^2}.$$

Therefore,

$$\boxed{1 - 8\mathrm{e}^{-m_0(\Delta)} = O(\Delta^2).} \tag{40}$$

This implies

$$m_0(\Delta) \to \ln 8^+.$$

Define the physical mass m by choosing

$$\boxed{1 - 8\mathrm{e}^{-m_0(\Delta)} = \mathrm{e}^{-m_0(\Delta)} m^2 \Delta^2.} \tag{41}$$

Multiplying by $\mathrm{e}^{m_0(\Delta)}$ gives

$$\mathrm{e}^{m_0(\Delta)} = 8 + m^2\Delta^2. \tag{42}$$

Thus

$$\boxed{m_0(\Delta) = \ln(8 + m^2\Delta^2) = \ln 8 + \frac{m^2\Delta^2}{8} + O(\Delta^4).} \tag{43}$$

14 Continuum Klein–Gordon Propagator

With this tuning,

$$D_\Delta(p) = \mathrm{e}^{-m_0(\Delta)}\Delta^2(p^2+m^2) + O(\Delta^4 p^4). \tag{44}$$

Therefore,

$$G_\Delta(x) \simeq \mathrm{e}^{m_0(\Delta)}\Delta^2 \int \frac{\mathrm{d}^4p}{(2\pi)^4}\,\frac{\mathrm{e}^{ip\cdot x}}{p^2+m^2}. \tag{45}$$

The raw path sum contains the overall lattice-dependent factor $\mathrm{e}^{m_0}\Delta^2$.
Remove it by defining

$$\boxed{D_E(x) = \lim_{\Delta\to 0}\frac{\mathrm{e}^{-m_0(\Delta)}}{\Delta^2}G_\Delta(x).} \tag{46}$$

Using Eq. (45),

$$\boxed{D_E(x) = \int_{\mathbb{R}^4}\frac{\mathrm{d}^4p}{(2\pi)^4}\,\frac{\mathrm{e}^{ip\cdot x}}{p^2+m^2}.} \tag{47}$$

This is the Euclidean propagator of a free Klein–Gordon field.

Final Result

The exact lattice propagator is

$$\boxed{G_\Delta(n) = \int_{[-\pi,\pi]^4}\frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{\mathrm{e}^{ik\cdot n}}{1-2\mathrm{e}^{-m_0}\sum_{\mu=1}^{4}\cos k_\mu}.} \tag{48}$$

The necessary tuning is

$$\boxed{m_0(\Delta) = \ln(8+m^2\Delta^2) = \ln 8 + \frac{m^2\Delta^2}{8} + O(\Delta^4).} \tag{49}$$

Finally,

$$\boxed{D_E(x) = \lim_{\Delta\to 0}\frac{\mathrm{e}^{-m_0(\Delta)}}{\Delta^2}G_\Delta(x) = \int_{\mathbb{R}^4}\frac{\mathrm{d}^4p}{(2\pi)^4}\,\frac{\mathrm{e}^{ip\cdot x}}{p^2+m^2}.} \tag{50}$$

AI Use Declaration

Model(s) used: Perplexity AI (for generating the full LaTeX code and helping to explain the solution steps).

Revisions/corrections made to model output, and clarifications added:

I did not change the mathematical steps or the final formulas, because the derivation produced by the AI looked correct to me. My follow-up questions were about conceptual points that I did not fully understand at first: why we Fourier transform with respect to the lattice coordinate n instead of the path length L, why a momentum variable p appears when we set $k = \Delta p$ and try to match a continuum plane wave $e^{ip\cdot x}$, and how the tuning of $m_0(\Delta)$ is chosen so that the lattice result reproduces the known form of the continuum Klein–Gordon propagator.

The AI explained that the introduction of p is motivated by the fact that we already know what the Klein–Gordon propagator looks like in momentum space and are effectively designing the lattice limit to match it. Based on these answers, I only refined the textual explanations around these points; the equations and their order were left unchanged.

Comparisons/checks against the lecture notes:

I did not formally compare the output against the lecture notes, but since this material was covered in the previous lecture, I checked the main steps against what I remembered from class.

Calculation steps, limiting cases, or consistency checks verified by hand:

I reviewed and checked almost all main calculations on paper.

Calculation steps, limiting cases, or consistency checks subjected to automated verification (via Mathematica, Python, LLMs):

I did not check any calculation with Mathematica or Python; I only used a text-based LLM to generate and organize the LaTeX code and explanations.